Shahid Beheshti University

Department of Computer Science

Advanced Data Mining

Assignment 1

Behavioral Modeling and Recommendation Systems

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1 Theoretical Questions

Exercise 1 (The Geometry of Regularization: Ridge vs. Lasso)

When we have highly complex datasets—especially those with severe multicollinearity or more features than observations ($p > n$)—standard Linear Regression can perform poorly. Shrinkage methods such as Ridge (L2) and Lasso (L1) address this by penalizing coefficient magnitude.

a) Explain what happens to the model coefficients as the regularization parameter λ increases in Ridge and in Lasso. Why does shrinking coefficients often improve performance on unseen test data?

b) Describe the key geometric difference between the Ridge (ℓ_2) and Lasso (ℓ_1) penalties. Why does Lasso tend to produce exact zero coefficients (feature selection), while Ridge typically does not?

c) If several predictors are highly correlated, how does Ridge distribute their coefficients compared to Lasso? Which method would you choose if model simplicity and interpretability are the main goals, and why?

Exercise 2 (Classification Under Severe Class Imbalance)

Consider a highly imbalanced fraud detection problem.

a) Explain the trade-off between Sensitivity (True Positive Rate) and Specificity (True Negative Rate) as the classification threshold moves from 0 to 1.

b) Interpret what the Area Under the ROC Curve (AUC) represents about a model's ability to rank predictions, independent of any specific threshold.

c) Describe how the ROC curve changes if the model performs no better than random guessing. (Hint: consider how well the predicted probability distributions of the two classes are separated.)

Exercise 3 (Support Vector Machines and Robustness to Outliers)

In a linearly separable 2D classification problem, suppose a single noisy data point is incorrectly placed inside the opposite class cluster.

a) Explain how a pure Maximal Margin Classifier would adjust its decision boundary and margin in response to this single outlier, and why this can lead to poor generalization.

b) Contrast this with a Support Vector Classifier (soft margin) and describe how it geometrically handles such noisy points.

c) Discuss the role of the regularization parameter C:

1) What happens to the margin width?
2) How many margin violations are allowed?
3) How does the bias–variance trade-off change when C is very small versus very large?

(Hint: consider whether a strict or flexible margin is more prone to overfitting noise.)

Exercise 4 (Resampling and the Variance of Hold-Out Validation)

When evaluating a complex diagnostic model using a single 50/50 train–validation split:

a) Explain why different random splits can produce drastically different performance estimates. What does this reveal about the high variance and instability of the validation-set approach?

b) Since the model is trained on only half of the available data, how does this affect the reliability of the estimated test error? Does this method tend to overestimate or underestimate real-world performance, and why?

c) What additional problems arise if (i) the target variable is highly imbalanced (e.g., 1% positive cases), or (ii) the dataset contains extreme outliers? (Hint: consider learning curves and what happens when important or rare observations are unevenly distributed across splits.)

Exercise 5 (Tree-Based Classification: Impurity Measures)

When building a classification tree, splits are evaluated using metrics such as Gini Index, Entropy, or Classification Error.

a) What does it mean for a node (region of feature space) to be *pure*? Why is maximizing node purity the main objective when growing a classification tree?

b) Although Classification Error seems intuitive, why is it not preferred for growing trees? How do Gini Index and Entropy differ conceptually, and why is Classification Error too insensitive to detect meaningful improvements in purity?

c) While Gini and Entropy often produce similar results, why might Gini be preferred for very large datasets from a computational efficiency perspective? (Hint: consider whether classification error changes when class proportions improve but the majority class remains the same.)

Exercise 6 (Unsupervised Learning: PCA Geometry and Limitations)

a) What is the conceptual goal of PCA? Explain the geometric meaning of the first principal component in terms of the data cluster's shape and spread, without using mathematical formulas. How does it relate to the original feature axes?

b) If one feature is measured in millimeters and another in kilometers, what happens to the principal components if PCA is applied without standardization? Why does this undermine the purpose of PCA?

c) What is a key geometric limitation of standard PCA? If the data lies on a non-linear manifold (e.g., a horseshoe or Swiss roll), can PCA effectively reduce dimensionality while preserving true point relationships? Explain why or why not. (Hint: consider variance along stretched axes and whether a straight line can capture non-linear structure.)

2 Practical Questions

This component requires implementing a complete machine learning pipeline on a large-scale transactional dataset. Your work must reflect graduate-level experimental discipline, including principled feature design, strict avoidance of temporal leakage, and rigorous evaluation of both classification and ranking objectives. You must submit (i) a fully executable Python notebook containing all code, outputs, and explanatory markdown, and (ii) a technical report that documents methodology, results, and critical discussion. Any claim in the report should be supported by quantitative evidence (tables/figures) and clear interpretation.

Dataset: Instacart Market Basket Analysis

Use the Instacart Market Basket Analysis dataset from Kaggle:

https://www.kaggle.com/datasets/psparks/instacart-market-basket-analysis

You will primarily work with `orders.csv`, `order_products__prior.csv`, `order_products__train.csv`, `products.csv`, `aisles.csv`, and `departments.csv`. All preprocessing decisions, joins, and assumptions must be explicitly documented. In particular, feature construction must respect the temporal nature of purchase histories.

Phase I: Exploratory Data Analysis (EDA) and Feature Engineering

Your first objective is to characterize the dataset and extract behavioral regularities that motivate modeling decisions. EDA should quantify dataset scale and sparsity, describe user/product heterogeneity, and analyze the target variable `reordered` (including class imbalance). Visualizations must be clearly labeled and interpreted; plots presented without analytical conclusions will not receive full credit. Data quality issues (missingness, duplicates, anomalies/outliers) must be assessed and handled transparently.

EDA must include, at minimum:

- Descriptive statistics for users, products, and orders; dataset sparsity characterization.
- Distribution of orders per user; distribution of basket size.

- Global reorder rate analysis and class imbalance analysis for `reordered`.
- Product popularity and long-tail behavior; department/aisle distributions.
- Missing value handling and anomaly/outlier inspection; correlation analysis where appropriate.

Following EDA, you must design features at three levels: user-level, product-level, and user–product interaction level. Features should capture frequency, recency, intensity, and temporal dynamics of purchasing behavior. You are expected to justify feature choices conceptually and, where appropriate, empirically (e.g., ablation or feature importance analysis). **All features must be computed strictly using prior-order information for the relevant prediction point;** leakage is a critical methodological error.

Feature engineering must include (examples; you may add more):

- **User-level:** total number of orders; average basket size; reorder ratio; average time between orders; purchase frequency; recency statistics.
- **Product-level:** total purchase count; global reorder rate; popularity score; recency-related statistics.
- **User–product:** user-specific reorder rate; interaction frequency; recency of last purchase; product share within user history.

Phase II: Dimensionality Reduction and Behavioral Analysis

In this phase, you will examine whether user behavior admits a lower-dimensional structure that improves interpretability or reveals latent segments. Construct a user-level feature matrix and apply both linear and nonlinear techniques. The goal is not merely to generate embeddings, but to analyze representational geometry and compare methods with appropriate justification. Your discussion should highlight what structure is revealed (or not revealed) by each approach.

You must apply and compare:

- **PCA:** report explained variance ratio; provide cumulative explained variance; justify the number of components; interpret component loadings.
- **t-SNE and UMAP:** visualize embeddings; discuss clustering and structure preservation; comment on sensitivity to hyperparameters.

Phase III: Predictive Modeling (Reorder Prediction)

Reorder prediction is formulated as a binary classification task where the target is `reordered`. You must correctly merge the relevant tables to form a supervised learning dataset and provide a clear splitting strategy that avoids leakage (temporal and/or user-level, as appropriate). Modeling must focus on boosting-based methods, and you must implement and compare at least two distinct models. Comparisons must be supported by cross-validation, hyperparameter tuning, and careful metric interpretation.

Modeling constraints and requirements:

- Use boosting-based models such as Gradient Boosting, XGBoost, LightGBM, and/or CatBoost.
- Implement and compare at least **two** boosting models.
- Perform a minimum of **3-fold** cross-validation and systematic hyperparameter tuning.
- Report computational considerations (e.g., training time) and comment on trade-offs.

Evaluation metrics (must be reported and interpreted):

- AUC-ROC, Precision, Recall, F1-score, and confusion matrix.

Your analysis must address class imbalance, threshold selection strategy, and precision–recall trade-offs. Model selection must be justified empirically and discussed critically (not merely chosen by a single metric).

Phase IV: Top-K Recommendation System and Ranking Evaluation

Using predicted reorder probabilities, you will construct a Top-K recommender by ranking products for each user. You must justify the choice of K (fixed or adaptive) and clearly define the candidate set over which ranking is performed. The evaluation must use ranking metrics and should explicitly discuss how classification quality does (or does not) translate into ranking performance. A careful metric-based discussion is expected.

Recommendation evaluation metrics (required):

- Precision@K, Recall@K, Hit Rate@K, MAP@K, and NDCG@K.

Experimental Rigor and Reproducibility

All results must be reproducible. Fix random seeds where applicable, document data-processing steps, and structure the notebook as a clear pipeline (data loading $\rightarrow$ cleaning/joins $\rightarrow$ feature engineering $\rightarrow$ modeling $\rightarrow$ evaluation). The notebook should be readable as a scientific workflow rather than an unstructured log of trials. Provide sufficient detail so another researcher can reproduce your results from scratch.

Deliverables

Submit the following items:

- A single Jupyter notebook (`.ipynb`) containing all code, outputs, and explanatory markdown (fully executable).

- A technical report (PDF) written in formal academic style, including: introduction, dataset description, EDA findings, feature engineering methodology, dimensionality reduction comparison, modeling methodology and results, recommendation evaluation, and discussion of limitations and potential improvements.
- (Optional bonus) A deployment link (e.g., Streamlit/Hugging Face Spaces) that returns Top-K recommendations for a given `user_id`.